

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 3: Pteridophyta and Spermatophyta

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
D	B	B	D	B	D	D	A

USE AFTER: OBSERVE PHASE — Jigsaw, Part 1: Reproductive structures expert group examining the underside of the fern frond

Q1. A learner turns over a fern frond and sees rows of small brown dots on the underside. What are these brown dots?

- A. Flowers that have dried up
- B. Eggs laid by insects
- C. Seeds ready to be planted
- D. Sori, which hold spores for reproduction**

Correct answer: D. Brown dots on the underside of a fern frond are sori, which contain spores. Ferns reproduce by spores and make no seeds or flowers.

USE AFTER: OBSERVE PHASE — Jigsaw, Part 1: Roots expert group (fibrous vs tap root vs rhizome vs rhizoids)

Q2. A learner pulls up a maize plant and a bean plant and examines their roots. Which description is correct?

- A. Both have rhizoids instead of true roots
- B. Maize has fibrous roots and bean has a tap root**
- C. Both have a rhizome with fibrous roots
- D. Maize has a tap root and bean has fibrous roots

Correct answer: B. Maize has fibrous roots and bean has a tap root. Rhizoids belong to moss, and a rhizome with fibrous roots belongs to the fern.

USE AFTER: OBSERVE PHASE — Jigsaw, Part 1: Leaves expert group (counting veins: parallel or net)

Q3. A learner examines a leaf that is broad and has veins forming a net pattern. Which plant did the leaf most likely come from?

- A. Fern
- B. Bean plant**
- C. Maize plant
- D. Moss

Correct answer: B. Bean leaves are broad with net veins. Maize leaves are long and narrow with parallel veins, and ferns have fronds with pinnae.

USE AFTER: OBSERVE PHASE — Jigsaw, Part 2: home groups build the comparison table (Bryophyta | Pteridophyta | Spermatophyta)

Q4. A home group is completing its comparison table. Which feature separates a fern (Pteridophyta) from moss (Bryophyta)?

- A. The fern makes seeds but moss makes spores
- B. The fern has rhizoids only, but moss has true roots
- C. The fern has flowers but moss does not
- D. The fern has true roots and vascular tissue, but moss does not**

Correct answer: D. Ferns have true roots, stems, leaves (fronds) and vascular tissue. Moss lacks true roots and vascular tissue. Ferns do not make seeds or flowers.

USE AFTER: EXPLAIN PHASE — Gallery walk and class discussion: revisiting the prediction 'ferns have flowers'

Q5. A learner says, "Ferns are flowering plants, so they make seeds." Which evidence from the specimens shows this is wrong?

- A. A fern has no roots, so it cannot make seeds
- B. A fern frond has sori with spores on the underside, and it has no flowers or seeds**
- C. A fern frond is green, and only plants with other colours make seeds
- D. A fern frond is smaller than a bean plant, so it makes no seeds

Correct answer: B. Ferns reproduce by spores held in sori and do not have flowers or seeds. Ferns do have true roots, and size and colour are not used to decide how a plant reproduces.

USE AFTER: EXPLAIN PHASE — Notebook sentences: 'The feature that best separates a fern from a bean plant is ____ because ____'

Q6. Both ferns and bean plants have vascular tissue, true roots, stems and leaves. Which feature best separates a fern from a bean plant?

- A. A fern reproduces by seeds in pods, while a bean plant reproduces by spores
- B. A bean plant has rhizoids, but a fern has true roots
- C. A fern has vascular tissue, but a bean plant does not
- D. A fern reproduces by spores in sori, while a bean plant reproduces by seeds**

Correct answer: D. Both groups have vascular tissue and true roots, so these do not separate them. Ferns reproduce by spores and spermatophytes such as beans reproduce by seeds.

USE AFTER: EXPLAIN PHASE — Class discussion: 'Would a farmer in Kericho and a student in Kisumu both be able to check spores and seeds?'

Q7. A farmer in Kericho and a student in Kisumu both want to place a plant in the same group. Why is checking for spores or seeds better than grouping by size or colour?

- A. Spores and seeds are always bigger than the plant parts that show colour
- B. Only scientists are allowed to look for spores and seeds
- C. Size and colour never change, but spores and seeds do
- D. Anyone can observe and check spores or seeds in the same way, so people reach the same grouping**

Correct answer: D. Classification needs shared, observable features that anyone can check and agree on. Size, colour or use as food differ from person to person, which is why learners' own groupings disagreed.

USE AFTER: MODEL BUILDING PHASE — Revising the class model (flow chart): Vascular tissue? then Seeds? to place moss, fern, maize and bean

Q8. In the class flow chart, the first question is "Vascular tissue?" A plant answers "Yes", and the second question is "Seeds?" The plant answers "No, it makes spores". Which group does it belong to?

- A. Pteridophyta (fern)**
- B. Bryophyta (moss)
- C. Spermatophyta (maize, bean)
- D. It is not a plant, like the mushroom

Correct answer: A. A plant with vascular tissue that reproduces by spores and has no seeds is a pteridophyte, such as a fern. Moss has no vascular tissue, and spermatophytes make seeds.